Topic: Reducing Balance Loans **Name:** Solns

LOANS
There are **two types** of loans available for borrowing a sum of money, **flat rate loans** and **reducing rate loans** more commonly known as reducible interest loans.

Type 1: Flat rate loans

Flat rate loans
For a flat rate interest loan, the interest is calculated as simple interest for the full term of the loan and then this interest is spread equally over the number of repayments that are to be made to repay the loan.

Ex 1.

Jack borrows $6000 at a flat rate of 4.2% per annum. The loan is to be repaid monthly over two years.
Calculate: (a) the total interest paid on this loan. (b) the monthly repayments to be made.

a) $I = \frac{PRT}{100} = \frac{600 \times 4.2 \times 2}{100} = \$504$

$\therefore$ The total interest paid on this loan is $504.

b) Total amount borrowed = $6000 + $504 = $6504

$\therefore$ Monthly repayments $= \frac{\$6504}{24} = \$271$ / month

2 years = 2 × 12

Hence, 24 monthly repayments each of $271 need to be made.

Type 2: Reducing rate loans

Reducing rate loans
For a reducing rate loan, the interest is calculated on how much is still owing at the end of a specified period. The amount owed decreases as each payment is made, which means that less interest needs to be paid and hence it is known as a **reducible interest** loan. In other words, a reducible interest loan is one in which the regular repayments are greater than the interest charged for that period and hence by reducing the amount owed, future interest charges are reduced.

Ex 2.

× 0.0035

Jill borrows \$6000 at a reducible interest rate and agrees to repay the loan by making monthly repayments of \$265. The spreadsheet below summarises the progress of this loan over the twenty four month period.

	A	B	C	D	E
1	Month	Amount owed at the	Interest charge	Repayment	Amount owed at the
2		start of the month	for the month		end of the month
3	1	6000.00	21.00	265	5756.00
4	2	5756.00	20.15	265	5511.15
5	3	5511.15	19.29	265	5265.44
6	4	5265.44	18.43	265	5018.86
7	5	5018.86	17.57	265	4771.43
8	6	4771.43	16.70	265	4523.13
9	7	4523.13	15.83	265	4273.96
10	8	4273.96	14.96	265	4023.92
11	9	4023.92	14.08	265	3773.00
12	10	3773.00	13.21	265	3521.21
13	11	3521.21	12.32	265	3268.53
14	12	3268.53	11.44	265	3014.97
15	13	3014.97	10.55	265	2760.53
16	14	2760.53	9.66	265	2505.19
17	15	2505.19	8.77	265	2248.96
18	16	2248.96	7.87	265	1991.83
19	17	1991.83	6.97	265	1733.80
20	18	1733.80	6.07	265	1474.87
21	19	1474.87	5.16	265	1215.03
22	20	1215.03	4.25	265	954.28
23	21	954.28	3.34	265	692.62
24	22	692.62	2.42	265	430.05
25	23	430.05	1.51	265	166.55
26	24	166.55	0.58	265	-97.87
27		Total	262.13 **		
28					

below \$2000

NOTE: The entry in cell E26 is -97.87 which informs us that the loan has been **over-paid** by \$97.87.

(a) Calculate the annual interest rate that Jill must pay on this loan.
(b) Calculate the total interest paid on this loan.
(c) How long does it take for the balance of this loan to first fall below \$2000?

a) Monthly interest rate $= \frac{21}{6000} \times 100\% = 0.35\%$ per month.

$\Rightarrow$ Annual interest rate $= 0.35\% \times 12 = 4.2\%$ p.a.

b) Final payment (24th month) $= 166.55 + 0.58 = \$167.13$

Total of 23 repayments $= 23 \times 265 = \$6095$

Total of 24 repayments $= 6095 + 167.13 = \$6262.13$

Hence, total interest paid $= 6262.13 - 6000 = \$262.13$

borrowed

** Note: Total interest = \$262.13 from table.

c) It took <u>16 months</u> for the balance to fall below \$2000.

[Note: End of the 16th month $\Rightarrow$ \$1991.83]

NOTE: 1. Comparing the two loans discussed in examples 1 and 2 one can see the huge impact that is made on the total interest charges by a reducing rate loan. For the examples investigated we can see that the cost of the flat rate loan is nearly double that of the reducing rate loan.

2. The interest rate is given as 4.2% per annum and has to be altered to a monthly interest rate of $\frac{4.2\%}{12}$ = 0.35% per month to coincide with the repayment period.

3. The monthly interest charge may be calculated by using the formula:
Monthly interest charge = 0.35% of Amount owed at start of the month = $\frac{0.35}{100}$ x \$6000 = \$21.

4. The amount owed at the end of each month may be calculated using the formula:
Amount owing at end of the month = Amount owing at start + Interest – Repayment
For example: Amount owing at end of 2^{nd} month = \$5756 + \$20.15 – \$265 = \$5511.15.

5. A **recursive formula** may also be used to find either the amount owing at the start or end of the month: The recursive equation: $T_{n+1} = 1.0035T_n - 265,\ T_1 = 6000$ gives the amount owing at the start of the month. The same equation but with $T_1 = 5756$ gives the amount owing at the end of the month.

6. The spreadsheet above clearly shows the progressive reducing interest charges being generated by the repayment each month which is greater than the interest charged for that month.

7. The last repayment can be calculated by adding the negative amount (-\$97.87) in the amount owed at the end of the month column to the monthly repayment of \$265.
That is \$265 + (-\$97.87) = \$167.13.
The negative amount (-\$97.87) is the amount which has been paid in excess of the amount owed.
Alternatively add the amount owed at the start of the last month of the loan to the interest charge for the last month of the loan. That is \$166.55+\$0.58 = \$167.13.
Another alternative is to multiply the amount owed at the start of the last month of the loan by the growth factor that is 1.0035. In this case \$166.55 x 1.0035 = \$167.132925, i.e. \$167.13.

8. **Amortization** of a loan means paying off the loan and all interest charges owed on the loan by making fixed, regular repayments over a period of time.

* **Ex 3.**

Marcia borrowed $24 000 to purchase a new car. Interest on the loan was set at 7.5% per annum compounded monthly with monthly repayments of $600.

(a) Write a recursive rule to calculate the amount owing at the end of each month.
(b) How much does Marcia owe after two years?
(c) What is the amount of the final payment that Marcia will make?
(d) How much interest did Marcia pay for this loan?
(e) Marcia's friend suggests that doubling the repayments to $1200 per month will halve the interest charged. Is Marcia's friend correct? Justify your answer mathematically.

a) Monthly Interest $= \frac{7.5\%}{12} = 0.625\% = 0.00625$, as decimal

multiplier $= 1 + 0.00625 = 1.00625$

$$\begin{cases} T_{n+1} = 1.00625\,T_n - 600 \\ T_0 = 24000 \end{cases}$$

G.C.

sequence

$a_{n+1} = 1.00625\,a_n - 600$

$a_0 = 24000$

$\Rightarrow$ Find a_{24}

b) After 2 years $= 2\times 12 = 24$ months

$T_{24} = \$12386.97$

$\therefore$ Marcia owes \$12386.97 after 2 years.

* c) method 1 : $n = 46 \Rightarrow T_{46} = 103.315622279053 \approx \$103.32$

Final payment $= 1.00625 \times 103.315622279053 \approx \$103.96$

or method 2 : $n = 47 \Rightarrow T_{47} = -496.038655081703 \approx \$496.04$

Final payment $= \$600 - \$496.04 \approx \$103.96$

d) Total repaid $= \$600 \times 46 + 103.96 = \$27703.96$

Total interest $=$ Total repaid $-$ Amount borrowed

$= \$27703.96 - \$24000$

$= \$3703.96$

e) New recursive rule :

$$\begin{cases} T_{n+1} = 1.00625\,T_n - 1200 \\ T_0 = 2400 \end{cases}$$

$n = 22 \Rightarrow T_{22} = -681.0197 \approx -681.02$

Final payment $= 1200 - 681.02$

$= \$518.98$

Total repaid $= 1200 \times 21 + 518.98$

$= \$25718.98$

Total interest $= 25718.98 - 24000$

$= \$1718.98$

From (d) : half of interest $= \frac{3703.96}{2}$

$= \$1851.98$

Hence, Marcia's friend is wrong.

$\Rightarrow$ Double repayments gives interest payment $= \$1718.98$, which is less than half of \$3703.96.

Complete Ex 3A

Ex 4.

A sum of $60 000 was borrowed at an interest rate of 9% per annum. It was agreed that the loan had to be repaid by equal instalments over 10 years. The interest is calculated annually and the repayments are made annually over a set period of 10 years.

(a) Determine the annual repayment.

(b) Calculate, to the nearest cent, (i) the total amount that must be repaid to amortise the loan, and (ii) the amount of interest that is paid to amortise the loan.

(c) Verify the annual repayments using a the spreadsheet.

(d) During which year is the balance of the loan halved?

Graphic Calculator
Menu
↓
Financial
↓
Compound Interest

a) Using financial application (G.C.)

N = 10
I% = 9
PV = 60000
PMT = ? ⇒ PMT = −9349.205395 ≅ −$9349.21 [Negative number because the money is taken from us]
FV = 0
P/Y = 1
C/Y = 1

Hence, the annual repayment is $9349.21.

b) i) Total repayment = $9349.205395 × 10 ≈ $93492.05

ii) Interest paid = $93492.05 − $60000 = $33492.05

c) Spreadsheet:

Col A. YEAR	Col B. Opening Balance	Col. C Interest	Col. D Repayment	Col. E Closing Balance
1	60000	5400 ↑ =B1×0.09	9349.21	56050.79 ↑ =B1+C1−D1
2				
⋮				
10	8577.25	771.95	9349.21	0.00

∴ The spreadsheet verifies that the annual repayment of $9349.21 would make the loan is to be repaid after 10 repayments.

d) Year 7 → closing balance = $23665.59

The loan is halved during the 7th year.

SOLUTION

(a) As there has been no indication that a flat rate of interest applies we automatically assume that we are dealing with a reducing balance loan.

In order to find the annual repayment the financial program of a calculator may be used.

For our situation:

N = 10 as we are making annual instalments for 10 years.
I% = 9 because the annual interest rate is 9%.
PV = 60000 is the value of the loan.
PMT = unknown as it is the amount of the instalment.
FV = 0 as the loan will be paid off and its future value will be $0.
P/Y = 1 as we are making 1 instalment per year.
C/Y = 1 as we are compounding the interest once per year.

Compound Interest

N	10
I%	9
PV	60000
PMT	-9349.205395
FV	0
P/ Y	1
C/ Y	1

On entering the given parameters and tapping the variable that we wish to determine, in this case the PMT, the calculator display is completed. The calculator shows that PMT = -9349.205395 which informs us that we must make 10 annual payments of $9349.21 to repay the loan under the agreed conditions. The repayments are displayed as a **negative** number because it is money that is outgoing, i.e. money that is taken from us. The PV amount on the other-hand is **positive** as it is money that is incoming, i.e. money that has been given to us.

(b) (i) Total repayment = Annual repayment x Number of repayments
= $9349.205395 x 10
= $93492.05395

Total amount to be repaid is $93492.05

(ii) Interest paid = Total repaid – Amount borrowed
= $93492.05 – $60000
= $33492.05

(c)

	A	B	C	D	E
1	Year	Opening Balance	Interest	Repayment	Closing balance
2	1	60000.00	5400.00	9349.21	56050.79
3	2	56050.79	5044.57	9349.21	51746.16
4	3	51746.16	4657.15	9349.21	47054.11
5	4	47054.11	4234.87	9349.21	41939.77
6	5	41939.77	3774.58	9349.21	36365.15
7	6	36365.15	3272.86	9349.21	30288.81
8	7	30288.81	2725.99	9349.21	23665.59
9	8	23665.59	2129.90	9349.21	16446.29
10	9	16446.29	1480.17	9349.21	8577.25
11	10	8577.25	771.95	9349.21	0.00

The spreadsheet verifies the annual repayments because the loan is repaid after 10 repayments of $9349.21.

(d) Row 8 of the spreadsheet indicates that the starting balance at the beginning of the 7^{th} year is $30288.81 with a closing balance of $23 665.59. This tells us that the loan is halved during the 7^{th} year.

Ex 5.

At the beginning of one month \$1000 is invested into an account paying interest at 7·5% per annum, compounded monthly, and an extra \$100 is invested at the end of that first month and the end of every month thereafter. How much is the account worth at the end of the month two years later, just after the \$100 deposit for that month is made? (For the purposes of this question assume that a year consists of 12 equal months.)

monthly interest $= \frac{7.5\%}{12} = 0.625\% = 0.00625$, as decimal

multiplier $= 1 + 0.00625 = 1.00625$

Method 1

$$\begin{cases} T_{n+1} = 1.00625 \times T_n + 100 \\ T_0 = 1000 \end{cases}$$

G.C.

$$\begin{cases} a_{n+1} = 1.00625 a_n + 100 \\ a_0 = 1000 \end{cases}$$

Value at end of 24th month $= T_{24} = \$3741.96$ (ie 2 years)

Hence, the account will be worth \$3741.96.

Method 2 "Financial"

N = 24
I = 7.5
PV = −1000
Pmt = −100
FV = ?? ⇒ 3741.964308
P/Y = 12
C/Y = 12

Ex 6.

Johan takes out a loan of \$7000. Interest of 9·4% per annum is added to the loan annually and Johan repays \$800 at the end of each one year period. How much will Johan still owe immediately after he makes the \$800 repayment at the end of year 8?

Method 1 Multiplier $= 100\% + 9.4\% = 109.4\% = 1.094$, as decimal.

$$\begin{cases} T_{n+1} = 1.094 T_n - 800 \\ T_0 = 7000 \end{cases}$$

$\Rightarrow T_8 = \$5411.09$

Hence, the amount still owed at the end of year 8 is \$5411.09.

Method 2

N = 8
I% = 9.4
PV = 7000 **
PMT = −800
FV = ?? ⇒ −5411.085299
P/Y = 1
C/Y = 1

Hence, the amount still owed \$5411.09

Ex 7.

A sum of $60 000 was borrowed at an interest rate of 9% per annum. It was agreed that the loan had to be repaid by equal instalments over 10 years. The interest is calculated monthly and repayments are made monthly.

(a) Determine the monthly repayment.

(b) Calculate, to the nearest cent, the total amount that must be repaid to amortise the loan.

a) Using "Financial" application

N = 120

I = 9

PV = 60000

PMT = ?? ⇒ -760.0546425

≈ -760.05

FV = 0

P/Y = 12

C/Y = 12

Hence, the monthly repayment is ≈ $760.05

b) Total repayment

10 years x12

= 760.0546425 x 120

= 91206.5568

≈ $ 91206.56

Ex 8.

A sum of $25 000 is borrowed at a rate of 3.6% per annum compounded monthly with monthly repayments of $460.

(a) How much is owing on this loan after 3 years?

(b) How many repayments need to be made to fully repay this loan?

a) Using "Financial" Application

N = 36

I% = 3.6

PV = 25000

PMT = -460

*

FV = ?? ⇒ -10386.98564

P/Y = 12

C/Y = 12

Hence, the amount still owed after 3 years is $10386.99

b)

N = ?? ⇒ 59.41666555

I% = 3.6

PV = 25000

PMT = -460

FV = 0

P/Y = 12

C/Y = 12

Hence, 60 repayments need to be made with the last repayment less than $460.

Complete Ex 3B

Extension: Using the "Amortization" Application from a graphic calculator

Remember that **amortization** of a loan means paying off the loan and all interest charges owed on the loan by making fixed, regular repayments over a period of time.

From Ex 4: Another Approach

A sum of \$60000 was borrowed at an interest rate of 9% per annum. It was agreed that the loan had to be repaid by equal instalments over 10 years.

Situation 1:
The interest is calculated annually and the repayments are made annually over a set period of 10 years.

(a) Determine the annual repayment.
As there has been no indication that a flat rate of interest applies we automatically assume that we are dealing with a reducing balance loan.
In order to find the annual repayment the financial program of a calculator may be used.

For our situation:

N = 10 as we are making annual instalments for 10 years.
I% = 9 because the annual interest rate is 9%.
PV = 60000 is the value of the loan.
PMT = unknown as it is the amount of the instalment.
FV = 0 as the loan will be paid off and its future value will be \$0.
P/Y = 1 as we are making 1 instalment per year.
C/Y = 1 as we are compounding the interest once per year.

On entering the given parameters and tapping the variable that we wish to determine, in this case the PMT, the calculator display is completed
The calculator shows that PMT = -9349.205395 which informs us that we must make 10 annual payments of \$9349.21 to repay the loan under the agreed conditions. The repayments are displayed as a negative number because this is money that we have to pay, that is, it is money outgoing.

Compound Interest

N	10
I%	9
PV	60000
PMT	-9349.205395
FV	0
P/ Y	1
C/ Y	1

(b) Calculate, to the nearest cent, (i) the total amount that must be repaid to amortise the loan, and (ii) the amount of interest that is paid to amortise the loan.

The answers may be found as follows:

(i) Total repayment = Annual repayment x Number of repayments
= \$9349.205395 x 10
= \$93492.05395

Total amount to be repaid is \$93492.05

(ii) Interest paid = Total repaid – Amount borrowed
= \$93492.05 – \$60000
= \$33492.05

The answers to (b) and other questions may be found using the **Amortization** application of the calculator.
This is accessed by tapping the Calculation menu.
This window enables us to access more information about the loan over its life-span of ten years.

It is important to note that the details of this loan entered above are carried over to the Amortization option.
The carried forward details (in bold font) which are displayed are :

- the annual interest rate of 9%
- PV, the amount of the loan
- PMT, the amount of each instalment or payment.
- P/Y, the number of instalment periods per year
- C/Y, the number of times the interest is compounded per year

Amortization

PM1	
PM2	
I%	9
PV	60000
PMT	-9349.205395
P/ Y	1
C/ Y	1
BAL	
INT	
PRN	
Σ INT	
Σ PRN	

Note:

Making an entry in PM1, for example, 3, instructs the program that 3 is the number of the first instalment or payment in the period being considered. An entry in PM2, for example, 8, instructs the program that the last payment or instalment in the period under consideration is the eighth .

The display for the command $\boxed{\text{BAL}}$ gives the balance of the principal, that is the amount still owing after the payment number entered in PM2.
If we enter 1 in PM1 and 2 in PM2 and $\boxed{\text{BAL}}$ is tapped the display is 51746.16073 which informs us that after the second payment on the loan is made $51 746.16 is still outstanding. Note that the displayed balance is positive, informing us that is money that we have. Verify the balance of $51 746.16 using your calculator and check that the answer is correct by referring to the spread sheet found below.

The display for the command $\boxed{\text{INT}}$ gives the **interest portion** of any payment or instalment number that is entered in PM1.
If 4 is entered in PM1 and $\boxed{\text{INT}}$ tapped the display is -4234.869882, checking with the spreadsheet below verifies that the interest accrued during the 4^{th} year is $4234.87 and hence $4234.87 of the repayment of $9349.21 will be utilised to pay off the interest accrued during the 4^{th} year. The negative sign reminds us that this interest is a part of the PMT entry of -9349.205395.

The display for the command $\boxed{\text{PRN}}$ gives the **principal portion** of any payment or instalment number that is entered in PM1.
If 4 is entered in PM1 and $\boxed{\text{PRN}}$ tapped, -5114.335513 is displayed. This informs us that $5114.34 of the 4^{th} repayment will be utilised to reduce the outstanding principle of the loan as the remainder, $9349.21 – $5114.34 = $4234.87 has been used to pay the interest charged for the 4^{th} year. The negative sign reminds us that this amount is a part of the PMT entry of -9349.205395.
Summing up, the 4^{th} repayment of -9349.205395 has been divided into -4234.869882 which nullifies the interest charge for the 4^{th} year and -5114.335513 which reduces the amount owed.

The display for the command $\boxed{\Sigma\text{INT}}$ gives total interest paid from PM1 to PM2 inclusive.
If 2 is entered in PM1 and 4 in PM2 and $\boxed{\Sigma\text{INT}}$ tapped, -13936.59586 is displayed. This informs us that the total interest paid from payment 2 to 4 inclusive was $13936.60 which can be verified using the spreadsheet below. The negative sign reminds us that this is an amount of the repayment.

The display for the command $\boxed{\Sigma\text{PRN}}$ gives the total principal paid from PM1 to PM2 inclusive.
If 2 is entered in PM1 and 4 in PM2 and $\boxed{\Sigma\text{PRN}}$ tapped, -14111.02032 is displayed. This informs us that the outstanding principle has been reduced by $14111.02 between payments 2 to 4 inclusive. Using the spreadsheet below verifies this, that is 56050.79 – 41939.77 = 14111.02. The negative sign reminds us that this is a partial sum of the repayments.

(c) Verify the annual repayments using a spreadsheet

	A	B	C	D	E
1	Year	Opening Balance	Interest	Repayment	Closing balance
2	1	60000.00	5400.00	9349.21	56050.79
3	2	56050.79	5044.57	9349.21	51746.16
4	3	51746.16	4657.15	9349.21	47054.11
5	4	47054.11	4234.87	9349.21	41939.77
6	5	41939.77	3774.58	9349.21	36365.15
7	6	36365.15	3272.86	9349.21	30288.81
8	7	30288.81	2725.99	9349.21	23665.59
9	8	23665.59	2129.90	9349.21	16446.29
10	9	16446.29	1480.17	9349.21	8577.25
11	10	8577.25	771.95	9349.21	0.00

The spreadsheet verifies the annual repayments because the loan is repaid after 10 repayments of \$9349.21.

(d) What formulae have been used to calculate the display in the following cells?
(i) Cell C2 (ii) Cell E2 (iii) Cell B3 (iv) Cell D3

(i) Cell C2 = 0.09 xB2
(ii) Cell E2 = B2 + C2 – D2
(iii) Cell B3 =E2
(iv) Cell D3 = \$D\$2

(e) During which year is the balance of the loan halved?
Row 8 of the spreadsheet indicates that the starting balance at the beginning of the 7^{th} year is \$30288.81 with a closing balance of \$23665.59. This tells us that the loan is halved during the 7^{th} year.

Situation 2: The interest is calculated monthly and repayments are made monthly.

(a) Determine the monthly repayment.
Using the built in financial program of the calculator the monthly repayments are set at \$760.05.

Compound Interest

N	120
I%	9
PV	60000
PMT	-760.0546425
FV	0
P/ Y	12
C/ Y	12

(b) What changes need to be made to the spreadsheet used in situation 1 to adapt it for situation 2?
(i) Cell C2 must contain the formula = 0.09/12*B2
(ii) Fill down all cells to at least 121.
(iii) Cell D2 must be changed to 760.0546425.

	A	B	C	D	E
1	Month	Opening Balance	Interest	Repayment	Closing balance
2	1	60000.00	450.00	760.0546425	59689.95
3	2	59689.95	447.67	760.0546425	59377.57
4	3	59377.57	445.33	760.0546425	59062.84
5	4	59062.84	442.97	760.0546425	58745.76
6	5	58745.76	440.59	760.0546425	58426.30
7	6	58426.30	438.20	760.0546425	58104.44
8	7	58104.44	435.78	760.0546425	57780.17
9	8	57780.17	433.35	760.0546425	57453.47
10	9	57453.47	430.90	760.0546425	57124.31
11	10	57124.31	428.43	760.0546425	56792.69
12	11	56792.69	425.95	760.0546425	56458.58
13	12	56458.58	423.44	760.0546425	56121.96
14	13	56121.96	420.91	760.0546425	55782.82

* Note: Opening balance, interest and closing balance have been rounded to the nearest cent.

(c) Calculate, to the nearest cent, the total amount that must be repaid to amortize the loan.

Total repayment = Monthly repayment x Number of repayments
= \$760.0546425 x 120
= \$91206.5568

Total amount to be repaid is \$91206.56

The total amount that must be repaid to amortize the loan can be found using the Amortization application by setting PM1 to 1, PM2 to 120 and then by adding together $\boxed{\Sigma \text{INT}}$ and $\boxed{\Sigma \text{PRN}}$.
That is -31206.5571 + (-60000) = -91206.5571.

Amortization

PM1	1
PM2	120
I%	9
PV	60000
PMT	-760.0546425
P/ Y	12
C/ Y	12
BAL	
INT	
PRN	
Σ INT	-31206.5571
Σ PRN	-60000

(d) Determine the amount owing after the first year.

Using the amortization application by setting PM1 to 1 and PM2 to 12 gives \$56 121. 96 for command $\boxed{\text{BAL}}$.
This can be verified by checking the entry E13 of the spreadsheet.

(e) Calculate the interest charged during the first two months of the loan.

Using the amortization application gives -897.6745902.
That is, interest charged for the first two months of the loan is \$897.67. This can be readily checked by summing cells C2 and C3 of the spreadsheet.

NOTE: The financial programme used above is limited to situations where the repayments are all equal over the entire period of the loan.

Ex 9. [Ex3C, Q6]

A sum of $9000 was borrowed at an interest rate of 6.9% per annum. It was agreed that the loan had to be repaid by equal instalments over 2 years.
The interest is calculated monthly and repayments are made monthly.

(a) Determine the monthly repayment.

(b) Calculate, to the nearest cent, the total amount that must be repaid to amortise the loan.

(c) How much interest was paid on the life of the loan?

(d) What is the balance of this loan halfway through the loan period?

a) Using "Financial" application

N = 24

I% = 6.9

PV = 9000

PMT = ?? ⇒ −402.5453281

FV = 0

P/Y = 12

C/Y = 12

Hence, the monthly repayment is \$ 402.55

b) Total amount to be repaid

= 402.5453281 x 24

= 9661.087874 ≈ \$9661.09

c) Total interest paid

= 9661.09 − 9000

= \$ 661.09

d) Halfway through the loan period

⇒ n = 12 months

N = 12

I% = 6.9

PV = 9000

PMT = −402.5453281

FV = ?? ⇒ −4654.744321

P/Y = 12

C/Y = 12

Hence, the balance is \$4654.74

Ex 10. [Ex3C, Q13]

Ronald is repaying a $457 000 loan over 15 years with quarterly instalments at 7.14% per annum adjusted quarterly.

(a) Find the amount of each quarterly repayment.

(b) Calculate the total amount repaid on the loan.

(c) Calculate the interest paid on the loan.

(d) What is the balance owing after five years?

a) Using "financial"

N = 60 ⇐ 15 × 4

I% = 7.14

PV = 457000

PMT = ?? ⇒ −12471.60112

FV = 0

P/Y = 4

C/Y = 4

Hence, quarterly repayment of

$12471.60

b) Total repaid

= 12471.60112 × 60

= $748296.0672

≈ $748296.07

c) Total interest paid

= 748296.0672 − 457000

= 291296.0672

≈ $291296.07

d)

N = 20 ← 5 × 4

I% = 7.14

PV = 457000

PMT = −12471.60112

FV = ?? ⇒ −354392.5234

P/Y = 4

C/Y = 4

Hence, the balance owing after 5 years is $354392.52

Complete Ex 3C

Comparing Loans

We need to interpret, compare and make decisions about flat rate and reducing balance interest loans.

$\frac{5.7}{100} = 0.057$; $\frac{0.057}{12} = 0.00475$ in decimal

Ex 11. [Ex3D, Q1]

Which, and by how much is the better reducing balance interest loan?

LOAN A: $8 000 borrowed at 5.7% pa with monthly repayments of $570?

LOAN B: $8 000 borrowed at 7.5% pa with monthly repayments of $750?

Show all working to justify your response.

$\frac{7.5}{100} = 0.075$; $\frac{0.075}{12} = 0.00625$

* Loan A :

"Financial"

$\Rightarrow N = 14.55928424$ (15 repayments in total)

i.e. 14 full repayments + last payment

$= 14 \times 570 + (570 - 250.88) = \$8299.12$ ↖ use N = 15

$\Rightarrow$ Interest $= 8299.12 - 8000 = \$299.12$

OR $\begin{cases} A_{n+1} = 1.00475 A_n - 570 \\ A_0 = 8000 \end{cases}$

Total repayments $= 14 \times 570 + (570 - 250.88)$

$= 8299.12$

$\Rightarrow$ Interest $= 8299.12 - 8000 = \$299.12$

Loan B: $\Rightarrow N = 11.07332005$

"Financial" (11 full repayments + last payment)

$= 11 \times 750 + (750 - 694.85) = \$8305.15$

$\Rightarrow$ Interest $= 8305.15 - 8000 = \$305.15$

OR $\begin{cases} B_{n+1} = 1.00625 B_n - 750 \\ B_0 = 8000 \end{cases}$

Hence, loan A is a better loan (i.e. cheaper interest)

by $305.15 - 299.12 = \$6.03$

Ex 12. [Ex3D, Q9]

A sum of $100000 was borrowed at an interest rate of 12% per annum. It was agreed that the loan had to be repaid by equal instalments over 20 years. The repayment options were as follows:

OPTION A : annual repayments of $13387.88

OPTION B : quarterly repayments of $3311.17

OPTION C : monthly repayments of $1101.09

} rounding to the nearest cent $\Rightarrow$ minor error !

(a) Calculate(to the nearest cent) for each option the total amount that must be paid back to pay off the loan.

option A : Total $= 13387.88 \times 20 = \$267,757.60$

option B : Total $= 3311.17 \times 20 \times 4 = \$264,893.60$

option C : Total $= 1101.09 \times 20 \times 12 = \$264,261.60$

(b) Which option is the cheapest? (i.e. the least repayment)

$\Rightarrow$ Option C is the cheapest.

(c) How much is the saving between the dearest and cheapest options?

$$\text{Difference} = 267757.60 - 264261.60$$
$$= \$\,3496$$

(d) Write a recursive formula that will enable you to find the amount owing at the end of any quarter for option B.

quarterly interest $= \frac{12\%}{4} = 3\% = 0.03$ (decimal)

multiplier $= 1 + 0.03 = 1.03$

$$B_{n+1} = 1.03\,B_n - 3311.17$$
$$B_0 = 100\,000$$

(e) For option B how much is owing after (i) 2 years? (ii) 10 years? (iii) 20 years?

i) 2 years $\Rightarrow T_8 = \$97232.97$
(2×4)

ii) 10 years $\Rightarrow T_{40} = \$76537.39$
(10×4)

iii) 20 years $\Rightarrow T_{80} = \$1.47$
(20×4)

(f) Explain why the amount owing after 20 years is not zero.

Due to the rounding error (2dp) in the amount being repaid each quarter. The loan is underpaid by \$1.47.

Complete Ex 3D

CHANGING THE TERMS OF A LOAN

Loans today usually have provision for changing of terms. Interest rates are usually variable instead of being fixed for the duration of the loan. The amount of the repayment may change which will result in the life of the loan being reduced or extended. Lump sums may be made which will affect the duration of the loan and the amount of interest paid. Changing the terms of a loan can reduce or extend the life of a loan and hence the overall cost of the loan.

Ex 13.

Adam borrows \$80 000 at 8.6% per annum with interest compounding monthly. The loan is to be repaid over 10 years with equal monthly instalments.

(a) Calculate the amount of each instalment to the nearest cent.

After 4 years the interest rate was reduced by 1% to 7.6% per annum.

(b) Calculate the new amount of each instalment, to the nearest cent, to repay the loan in the given time.

(c) How much, to the nearest dollar, does Adam save as a result of the cut in the interest rate?

a) Using "Financial"

N = 120 ← 10x12

I% = 8.6

PV = 80000

* PMT = ?? ⇒ -996.1692462

FV = 0

P/Y = 12

C/Y = 12

Hence, the amount of each instalment is \$ 996.17

b) After 4 years, we need to work out the amount still owing.

N = 48 ← 4 x 12

I% = 8.6

PV = 80000

PMT = -996.1692492

* FV = ?? ⇒ -55877.75058

P/Y = 12

C/Y = 12

⇒ The remaing 6 years of the loan:

N = 72 ← 6x12

I% = 7.6

PV = 55877.75085

* PMT = ?? ⇒ -968.8406506

FV = 0

P/Y = 12

C/Y = 12

Hence, the new monthly repayment is \$ 968.84

c)

option 1: repaid at 8.6% p.a for 10 years:

= 996.1692462x120 = \$119540.31

option 2: repaid 8.6% p.a for 4 years and 7.6% for 6 years:

= 996.1692462 x 48 + 968.8406506x72

= 47816.12382 + 69756.52684

= \$ 117572.65

Interest saved = 119540.31 − 117572.65

= \$1967.66

≈ \$1968 (nearest dollar)

Complete Ex 3E

* **Ex 14.** (Challenge)

Roxanne borrowed $150 000 at a rate of 9.1% per annum compounded monthly. She makes monthly repayments of $1715.50 to repay the loan.
(a) How long will it take Roxanne to repay the loan?
After 4 years Roxanne increases her monthly repayments to $1920.
(b) Calculate by how many months the length of the loan is reduced by this increase in repayments.
(c) How much will Roxanne save on this loan by increasing her repayments to $1920 after 4 years?

a) Using "Financial" application

N = ?? ⇒ 144.0007518
I% = 9.1
PV = 150000
PMT = −1715.50
FV = 0
P/Y = 12
C/Y = 12

Hence, it will take approximately 144 months (or $\frac{144}{12} = 12$ years) to repay the loan.

b) • We need to work out the amount still owing after 4 years.

N = 48 ← 4 x 12
I% = 9.1
PV = 150000
PMT = −1715.50
FV = ?? ⇒ −116684.3804
P/Y = 12
C/Y = 12

⇒ The amount still owing = $116684.38

• We need to work out how long it will take to repay $116684.38 with a monthly repayment of $1920:

N = ?? ⇒ 81.77468008
I% = 9.1
PV = 116684.3804
PMT = −1920
FV = 0
P/Y = 12
C/Y = 12

Hence, It will take 82 months to repay the remainder of the loan $116684.38.

Time difference = 144 − (4x12 + 82)
= 144 − 130 = 14 months

c) option 1: total repayments for 144 months:
$= 1715.50 \times 144 + 1.284796747 \times \left(1 + \frac{0.091}{12}\right)$
leftover
= 247032 + 1.294539789
= $ 247033.29

option 2: total repayments
$= 1715.50 \times 4 \times 12 + \left(\$1920 \times 81 + 1477.446835 \times \left(1 + \frac{0.091}{12}\right)\right)$
= 82344 + 155520 + 1488.65
= $239352.65

saving = 247033.29 − 239352.65
= $7680.64

Complete Ex 3F